ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice b.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, examples and neat sketches.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 C ✓ 1.1.3 A ✓ 1.1.4 B ✓ 1.1.5 D ✓ 1.1.6 A ✓ 1.1.7 C ✓ 1.1.8 C ✓ 1.1.9 A ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 A ✓ 1.3.2 J ✓ 1.3.3 H ✓ 1.3.4 G ✓ 1.3.5 E ✓ 1.3.6 I ✓ 1.3.7 C ✓ 1.3.8 B ✓ 1.3.9 F ✓ 1.3.10 D ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1.1 $A = \pi d^2 \div 4 = 3{,}142 \times 0{,}08^2 \div 4$ ✓ $= 5{,}027 \times 10^{-3}\ m^2$ ✓; $F = p \times A$ ✓ $= 10 \times 10^6 \times 5{,}027 \times 10^{-3} = 50\ 272\ N\ (50{,}27\ kN)$ ✓ (4)

2.1.2 $\text{Rod area} = 3{,}142 \times 0{,}04^2 \div 4 = 1{,}257 \times 10^{-3}\ m^2$ ✓; $\text{annulus area} = 5{,}027 \times 10^{-3} - 1{,}257 \times 10^{-3} = 3{,}770 \times 10^{-3}\ m^2$ ✓; $F = p \times A$ ✓ $= 10 \times 10^6 \times 3{,}770 \times 10^{-3} = 37\ 704\ N\ (37{,}70\ kN)$ ✓ (4)

2.1.3 $Q = 20\ L/min = 0{,}02 \div 60 = 3{,}333 \times 10^{-4}\ m^3/s$ ✓✓; $v = Q \div A$ ✓ $= 3{,}333 \times 10^{-4} \div 5{,}027 \times 10^{-3} = 0{,}07\ m/s\ (66{,}3\ mm/s)$ ✓ (4)

2.2 Two lines (A and B) with steps 1 to 5 ✓; A rises at step 1–2 ✓; B rises at step 2–3 ✓; A falls at step 3–4 and B falls at step 4–5, back to the start ✓ (4)

Step	1	2	3	4	5 = 1
Cylinder A	in	out	out	in	in
Cylinder B	in	in	out	out	in

2.3 Hydraulics (any TWO) ✓✓: very large forces from small actuators; precise, stiff control of position because oil is nearly incompressible; smooth slow movement. Pneumatics (any TWO) ✓✓: clean (air leaks are harmless); fast; cheap and simple; safe in explosive areas; compliant / soft gripping (4)

[20]

QUESTION 3

3.1 P – output is proportional to the error; higher gain gives a faster response but can cause overshoot and oscillation, and leaves a steady-state error ✓✓; I – adds up the error over time and removes the steady-state error, but too much causes overshoot / slow oscillation ✓✓; D – reacts to the rate of change of error, damps the response and reduces overshoot, but is sensitive to noise ✓✓ (6)

3.2 Set point line ✓; output curve rising past the set point (overshoot labelled) ✓; rise time labelled ✓; settling time labelled where oscillation stays within the band ✓; steady-state error shown as the gap between the final value and the set point (or zero) ✓ (5)

3.3 $\text{Error} = 200 - 196 = 4 \text{ mm}$ ✓; $\%\text{ error} = 4 \div 200 \times 100$ ✓ $= 2\%$ ✓ (3)

3.4 Sudden starts and stops need very high acceleration, causing large forces, shock and vibration ✓; this can damage the mechanism, lose position or drop the part ✓; ramping speed up and down gives smooth, accurate movement and less wear ✓. (3)

[17]

QUESTION 4

4.1 Breakdown – repair only after the machine fails ✓✓; preventive – planned maintenance at fixed time or usage intervals to prevent failure ✓✓; predictive – the condition of the machine is measured and maintenance is done only when the measurements show it is needed ✓✓ (6)

4.2 Any THREE (technique ✓ + fault ✓): vibration analysis – unbalance, misalignment, bearing wear; thermography – hot spots, loose electrical connections, overloaded motors; oil analysis – wear particles, contamination; ultrasonic testing – air / gas leaks, bearing noise; motor current analysis – electrical / mechanical faults (6)

4.3.1 $\text{MTBF} = \text{operating time} \div \text{failures}$ ✓ $= 2\,000 \div 4 = 500 \text{ h}$ ✓ (2)

4.3.2 $\text{MTTR} = \text{repair time} \div \text{failures}$ ✓ $= 20 \div 4 = 5 \text{ h}$ ✓ (2)

4.3.3 $\text{A} = 500 \div (500 + 5) \times 100$ ✓ $= 99{,}01\%$ ✓ (2)

[18]

QUESTION 5

5.1.1 Horizontal axis: electric linear axis with stepper / servo motor and belt or ball screw ✓ – accurate positioning at several points ✓; vertical axis: pneumatic cylinder (or electric axis) ✓ – fast, simple up/down movement between two positions ✓; gripper: pneumatic parallel gripper or vacuum cup ✓ – quick, reliable gripping suited to the part ✓ (accept other motivated choices) (6)

5.1.2 Any FOUR with purpose ✓✓✓✓: part-present sensor at the feeder; home / limit sensors on the axes; cylinder end-position (reed) sensors; gripper closed / vacuum switch – confirms the part is held; fixture sensor – confirms the part was placed; safety light curtain / door switch (4)

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5.1.3 Any TWO with reason ✓✓: electrical / pneumatic circuit diagrams – for fault-finding; operating manual – safe operation; maintenance schedule – keeps it reliable; PLC program listing with comments – for changes; parts list – ordering spares; risk assessment / safety certificate (2)

5.1.4 The unit can be modelled and simulated on a computer before it is built ✓, so the program and movements can be tested and errors found without damaging equipment ✓, which shortens commissioning; later, live data can be compared with the model to detect problems ✓. (3)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE